THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE CHAPTER TEST SET B

Chapter 5: **Describing Motion Around Us** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A cyclist covers 300 m in 60 s along a straight road without turning back. The average speed is: 1

(a) 5 m s^{-1} **(b)** 0.2 m s^{-1} **(c)** 2.5 m s^{-1} **(d)** 10 m s^{-1}

2. A speed of 72 km h^{-1} in SI units is: 1

(a) 259.2 m s^{-1} **(b)** 24 m s^{-1} **(c)** 20 m s^{-1} **(d)** 40 m s^{-1}

3. The velocity of a car changes from 25 m s^{-1} to 5 m s^{-1} in 10 s in a straight line. Its average acceleration is: 1

(a) 2 m s^{-2} **(b)** −4 m s^{-2} **(c)** −2 m s^{-2} **(d)** 3 m s^{-2}

4. A body with initial velocity 20 m s^{-1} moves with constant acceleration -4 m s^{-2} for 3 s. Its final velocity is: 1

(a) 4 m s^{-1} **(b)** 8 m s^{-1} **(c)** 16 m s^{-1} **(d)** 48 m s^{-1}

5. A vehicle moving at 20 m s^{-1} is brought to rest with a uniform retardation of 5 m s^{-2}. The stopping distance is: 1

(a) 80 m **(b)** 40 m **(c)** 20 m **(d)** 4 m

6. A child on a merry-go-round of radius 14 m completes one full revolution ($\pi = 22/7$). The distance travelled and the displacement are: 1

(a) 44 m and 14 m **(b)** 88 m and 0 **(c)** 0 and 88 m **(d)** 88 m and 28 m

7. Average speed and the magnitude of average velocity are equal when an object: 1

(a) moves in a circle
(b) returns to start
(c) moves in a straight line in one direction
(d) is at rest

8. A straight line parallel to the time axis on a position–time graph shows that the object is: 1

(a) moving in a circle **(b)** moving with constant velocity
(c) at rest **(d)** accelerating

9. In uniform circular motion the velocity at any point is directed: 1

(a) along the radius **(b)** along the tangent
(c) towards the centre **(d)** away from the centre

10. A car goes from rest to 20 m s^{-1} in 10 s. The distance covered, assuming uniform acceleration, is: 1

(a) 20 m **(b)** 100 m **(c)** 200 m **(d)** 50 m

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 10–15 words each.

11. An athlete completes one round of a circular track of radius 7 m in 40 s. Find the average speed ($\pi = 22/7$). 2

12. Convert 90 km h^{-1} into m s^{-1}. 2

13. Define acceleration. What does a negative value indicate? 2

14. A car increases its speed from 10 m s^{-1} to 20 m s^{-1} in 5 s. Find its acceleration. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 30–40 words each.

15. State the characteristics of a position–time graph for 3
(i) an object at rest and
(ii) uniform motion.

16. Distinguish between the position–time graphs of uniform and non-uniform motion. 3

17. Draw a velocity–time graph for a body with constant acceleration and describe how acceleration is obtained. 3

OR

A velocity–time graph is a horizontal line at 20 m s^{-1} from 0 to 6 s. Find the acceleration and displacement. 3

18. A car moves with velocity rising uniformly from 5 m s^{-1} to 15 m s^{-1} in 10 s. Find the displacement using the graph. 3

OR

Derive $v = u + at$ using the definition of acceleration. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. (i) What does a velocity–time graph show for constant velocity and constant acceleration? 5
(ii) How is acceleration obtained from it?
(iii) Find the displacement for 5 m s^{-1} rising uniformly to 15 m s^{-1} in 10 s.

OR

Case study: Table: t = 0, 5, 10, 15, 20 s with v = 0, 2.5, 5, 7.5, 10 m s^{-1}. 5
(i) Describe the motion.
(ii) Find the acceleration.
(iii) Find the displacement in 20 s.
(iv) Describe the shape of the v–t graph.

20. (i) Derive $v = u + at$. 5
(ii) Derive $s = ut + \frac{1}{2}at^2$ using the v–t graph.
(iii) Derive $v^2 = u^2 + 2as$.

OR

A car starting from rest accelerates uniformly at 2 m s^{-2} for 10 s. 5
(i) Find its final velocity.
(ii) Find the distance in 10 s.
(iii) It then brakes to rest at 5 m s^{-2}; find the braking distance.
(iv) Find the total distance.

END OF QUESTION PAPER — Total: 40 Marks